Measuring Service Quality: A Reexamination and Extension
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J. Joseph Cronin, Jr. & Steven A. Taylor

Measuring Service Quality: A Reexamination and Extension

The authors investigate the conceptualization and measurement of service quality and the relationships between service quality, consumer satisfaction, and purchase intentions. A literature review suggests that the current operationalization of service quality confounds satisfaction and attitude. Hence, the authors test (1) an alternative method of operationalizing perceived service quality and (2) the significance of the relationships between service quality, consumer satisfaction, and purchase intentions. The results suggest that (1) a performance-based measure of service quality may be an improved means of measuring the service quality construct, (2) service quality is an antecedent of consumer satisfaction, (3) consumer satisfaction has a significant effect on purchase intentions, and (4) service quality has less effect on purchase intentions than does consumer satisfaction. Implications for managers and future research are discussed.

SERVICE industries are playing an increasingly important role in the overall economy of the United States (Bateson 1989; Ginzberg and Vojta 1981; Koepp 1987). In fact, the proportion of the U.S. population employed in the service sector increased from 30% in 1900 to 74% in 1984 (Bateson 1989). Koepp (1987) suggests that this sector is continuing to increase, as 85% of all the new jobs created since 1982 have been in service industries. Bateson (1989) further suggests that the growing importance of the service sector is not limited to the United States, as services currently account for 58% of the total worldwide GNP. There even appears to be executive consensus in the United States that service quality is one of the most important problems facing management today (Blackiston 1988; Cound 1988; Cravens 1988; Langevin 1988; Sherden 1988).

Interest in the measurement of service quality is thus understandably high and the delivery of higher levels of service quality is the strategy that is increasingly being offered as a key to service providers' efforts to position themselves more effectively in the marketplace (cf. Brown and Swartz 1989; Parasuraman, Zeithaml, and Berry 1988; Rudie and Wansley 1985; Thompson, DeSouza, and Gale 1985). However, the problem inherent in the implementation of such a strategy has been eloquently identified by several researchers: service quality is an elusive and abstract construct that is difficult to define and measure (Brown and Swartz 1989; Carman 1990; Crosby 1979; Garvin 1983; Parasuraman, Zeithaml, and Berry 1985, 1988; Rathmell 1966). In addition, to date the important relationships between service quality, customer satisfaction, and purchasing behavior remain largely unexplored.

Our research has two objectives. First, we suggest that the current conceptualization and operationalization of service quality (SERVQUAL) is inadequate. The SERVQUAL scale is based on Parasuraman, Zeithaml, and Berry's (1985) gap theory, which suggests that the difference between consumers' expectations about the performance of a general class of service providers and their assessment of the actual performance of a specific firm within that class drives the

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perception of service quality. However, little if any theoretical or empirical evidence supports the relevance of the expectations-performance gap as the basis for measuring service quality (Carman 1990). In fact, the marketing literature appears to offer considerable support for the superiority of simple performance-based measures of service quality (cf. Bolton and Drew 1991a,b; Churchill and Surprenant 1982; Mazis, Ahtola, and Klippel 1975; Woodruff, Cadotte, and Jenkins 1983). We therefore develop and test a performance-based alternative to the SERVQUAL measure.

The second objective is to examine the relationships between service quality, consumer satisfaction, and purchase intentions. Though these relationships have been discussed theoretically (cf. Bitner 1990; Bolton and Drew 1991a,b; Brown and Swartz 1989; Parasuraman, Zeithaml, and Berry 1988; Zeithaml, Parasuraman, and Berry 1990), they have not been subjected to a thorough empirical test. In particular, the purpose of the second phase of our study is to provide managers and researchers more information about (1) the causal order of the relationship between service quality and customer satisfaction and (2) the impact of service quality and customer satisfaction on purchase intentions. Simply stated, the managers of service providers need to know how to measure service quality, what aspects of a particular service best define its quality, and whether consumers actually purchase from firms that have the highest level of perceived service quality or from those with which they are most "satisfied."

After presenting theoretical background, we describe our research methods and results. We then discuss our findings and explore their implications for management and for future research. Finally, we examine the limitations of our study.

Theoretical Background

Service quality has been described as a form of attitude, related but not equivalent to satisfaction, that results from the comparison of expectations with performance (Bolton and Drew 1991a; Parasuraman, Zeithaml, and Berry 1988). A close examination of this definition suggests ambiguity between the definition and the conceptualization of service quality. Though researchers admit that the current measurement of consumers' perceptions of service quality closely conforms to the disconfirmation paradigm (Bitner 1990; Bolton and Drew 1991a), they also suggest that service quality and satisfaction are distinct constructs (Bitner 1990; Bolton and Drew 1991a,b; Parasuraman, Zeithaml, and Berry 1988). The most common explanation of the difference between the two is that perceived service quality is a form of attitude, a long-run overall evaluation, whereas satisfaction is a transaction-specific measure (Bitner 1990; Bolton and Drew 1991a; Parasuraman, Zeithaml, Berry 1988). Parasuraman, Zeithaml, and Berry (1988) further suggest that the difference lies in the way disconfirmation is operationalized. They state that in measuring perceived service quality the level of comparison is what a consumer *should* expect, whereas in measures of satisfaction the appropriate comparison is what a consumer *would* expect. However, such a differentiation appears to be inconsistent with Woodruff, Cadotte, and Jenkins' (1983) suggestion that expectations should be based on experience norms—what consumers *should* expect from a given service provider given their experience with that specific type of service organization.

Thus, the service literature has left confusion as to the relationship between consumer satisfaction and service quality. This distinction is important to managers and researchers alike because service providers need to know whether their objective should be to have consumers who are "satisfied" with their performance or to deliver the maximum level of "perceived service quality." The importance of this issue has led to several recent efforts to clarify the relationship between satisfaction and service quality (c.f., Bitner 1990; Bolton and Drew 1991a,b; Parasuraman, Zeithaml, and Berry 1985, 1988).

Initially Parasuraman, Zeithaml, and Berry (1985, 1988) proposed that higher levels of perceived service quality result in increased consumer satisfaction, but more recent evidence suggests that satisfaction is an antecedent of service quality (cf. Bitner 1990; Bolton and Drew 1991a,b). In particular, Bitner has demonstrated empirically a significant causal path between satisfaction and service quality in a structural equation analysis. In a second study, Bolton and Drew (1991a) used the common assumption that service quality is analogous to an attitude as a basis to suggest that satisfaction is an antecedent of service quality. Specifically, Bolton and Drew posit that perceived service quality ($ATTITUDE_t$) is a function of a consumer's residual perception of the service's quality from the prior period ($ATTITUDE_{t-1}$) and his or her level of (dis)satisfaction with the current level of service performance (CS/D_t[1]). This notion suggests that satisfaction is a distinct construct that mediates prior perceptions of service quality to form the current perception of service quality.

$$ATTITUDE_t = g(CS/D_t, ATTITUDE_{t-1}) \quad (1)$$

Bolton and Drew (1991a) indicate this relation implies that the disconfirmation process, expectations, and performance all should have a significant impact on

[1]CS/D_t = Consumer Satisfaction/Dissatisfaction.

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consumers' current perceptions of service quality. However, their results suggest that perceived service quality is strongly affected by current performance and that the impact of disconfirmation is relatively weak and transitory.

Finally, Bolton and Drew (1991b) extend the discussion of the relationship between satisfaction and service quality by proposing the following structural equations.

$$\text{Service Quality} = q_o\,(CS/D_t, \text{Disconfirmation}) \quad (2)$$

$$CS/D_t = c(\text{Disconfirmation, Expectations, Performance}) \quad (3)$$

To gain more insight into Bolton and Drew's findings, and into how service quality should be measured, we next briefly examine the satisfaction and attitude literatures.

Implications From the Satisfaction and Attitude Literatures

A major problem in the literature is the hesitancy to call perceived service quality an attitude. The literature's position is typified by Parasuraman, Zeithaml, and Berry's (1988) description of service quality as ". . . similar in many ways to an attitude" (p. 15). Researchers have attempted to differentiate service quality from consumer satisfaction, even while using the disconfirmation format to measure perceptions of service quality (cf. Bitner 1990; Carman 1990; Gronroos 1990; Heskett, Sasser, and Hart 1990; Parasuraman, Zeithaml, and Berry 1988; Zeithaml, Parasuraman, and Berry 1990). However, this approach is not consistent with the differentiation expressed between these constructs in the satisfaction and attitude literatures.

Oliver (1980) suggests that attitude (ATT) is initially a function of expectations (EXP) [ATT_{t1} = f(EXP)] and subsequently a function of the prior attitude toward and the present level of satisfaction (SAT) with a product or service [$ATT_{t2} = f(ATT_{t1}, SAT_{t2})$]. Purchase intentions (PI) then are considered initially to be a function of an individual's attitude toward a product or service [$PI_{t1} = f(ATT_{t1})$], but subject to modification due to the mediating effect on prior attitude of the satisfaction inherent in subsequent usages [$PI_{t2} = f(ATT_{t2}) = f(ATT_{t1}, SAT_{t2}$]. Thus, Oliver suggests that consumers form an attitude about a service provider on the basis of their prior expectations about the performance of the firm, and this attitude affects their intentions to purchase from that organization. This attitude then is modified by the level of (dis)satisfaction experienced by the consumer during subsequent encounters with the firm. The revised attitude becomes the relevant input for determining a consumer's current purchase intentions.

If one considers service quality to be an attitude, Oliver's (1980) study suggests that (1) in the absence of prior experience with a service provider, expectations initially define the level of perceived service quality, (2) upon the first experience with the service provider, the disconfirmation process leads to a revision in the initial level of perceived service quality, (3) subsequent experiences with the service provider will lead to further disconfirmation, which again modifies the level of perceived service quality, and (4) the redefined level of perceived service quality similarly modifies a consumer's purchase intentions toward that service provider.

Hence, Oliver's research suggests that service quality and consumer satisfaction are distinct constructs, but are related in that satisfaction mediates the effect of prior-period perceptions of service quality to cause a revised service quality perception to be formed. Satisfaction thus rapidly becomes part of the revised perception of service quality. This logic is consistent with Bolton and Drew's (1991a) findings and also calls into question the use of the disconfirmation framework as the primary measure of service quality, because disconfirmation appears only to mediate, not define, consumers' perceptions of service quality.

If in fact service quality is to be conceptualized as "similar to an attitude," perhaps more information could be generated for managers and researchers alike if the measurement of the construct conformed to an attitude-based conceptualization. A review of alternative attitude models suggests that the "adequacy-importance" form is the most efficient model to use if the objective is to predict behavioral intention or actual behavior (Mazis, Ahtola, Klippel 1975). In this model, an individual's attitude is defined by his or her importance-weighted evaluation of the performance of the specific dimensions of a product or service (see Cohen, Fishbein, and Ahtola 1972). However, experimental evidence indicates that the performance dimension alone predicts behavioral intentions and behavior at least as well as the complete model (Mazis, Ahtola, and Klippel 1975). This finding suggests using only performance perceptions as a measure of service quality.

A study by Churchill and Surprenant (1982) also partially supports the efficacy of using only performance perceptions to measure service quality. They conducted two experiments to examine the effects of expectations, performance, and disconfirmation on satisfaction. The results of one experiment suggested that performance alone determines the satisfaction of subjects. Woodruff, Cadotte, and Jenkins (1983) contribute additional support for performance-only measures of attitude. Again using the "adequacy-importance" model, they indicate that assimilation/contrast theory suggests that consumers may raise or

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lower their performance beliefs on the basis of how closely perceived performance approximates expected performance. Thus, they suggest that including importance weights and expectations only introduces redundancy. From the results of a field experiment, Bolton and Drew (1991a) also conclude that current performance ratings strongly affect attitudes whereas the effects of disconfirmation are generally insignificant and transitory. This study is particularly significant because the attitude examined is customers' perceptions of the quality inherent in a service.

Thus, the conclusion of the satisfaction and attitude literatures appears to be that (1) perceived service quality is best conceptualized as an attitude, (2) the "adequacy-importance" model is the most effective "attitude-based" operationalization of service quality, and (3) current performance adequately captures consumers' perceptions of the service quality offered by a specific service provider. In addition to the theoretical support for performance-based measures of service quality, practitioners often measure the determinants of overall satisfaction/perceived quality by having customers simply assess the performance of the company's business processes. Furthermore, the performance-based approach may actually be more in line with an antecedent/consequent conceptualization: that is, judgments of service quality and satisfaction appear to follow the evaluation of a service provider's performance. The first objective of our study is to examine these conclusions empirically by testing a performance-based measure of service quality as an alternative to the current disconfirmation-based SERVQUAL scale.

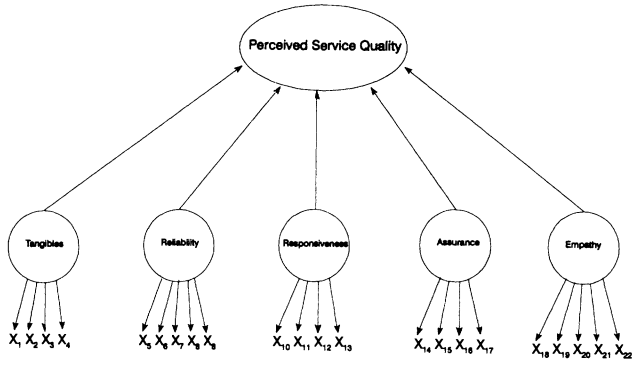
Operationalizing Service Quality

The current measurement of perceived service quality can be traced to the research of Parasuraman, Zeithaml, and Berry. These authors originally identified 10 determinants of service quality based on a series of focus group sessions (1985). They subsequently developed SERVQUAL (1988), which recasts the 10 determinants into five specific components: tangibles, reliability, responsiveness, assurance, and empathy (Figure 1).

The basis for identifying these five components was a factor analysis of the 22-item scale (see Appendix) developed from focus groups and from the specific industry applications undertaken by the authors (see Parasuraman, Zeithaml, and Berry 1985, 1988; and Zeithaml, Parasuraman, and Berry 1990 for a comprehensive review).

The scale development procedures employed appear to support the face validity of the 22 scale items (individual questions) included in the scale, but the issue of how the service quality measure should be

FIGURE 1
Service Quality as Conceptualized by Parasuraman, Zeithaml, and Berry (1988)



constructed and whether the individual scale items actually describe five separate service quality components is problematic. In fact, some empirical evidence suggests that the proposed delineation of the five components is not consistent when subjected to cross-sectional analysis (Carman 1990). Specifically, Carman found that some of the items did not load on the same component when compared across different types of service providers. However, though the veracity of conceptualizing the SERVQUAL scale as consisting of the five distinct components identified by Parasuraman, Zeithaml, and Berry (1988) has been questioned (Carman 1990), the validity of the 22 individual performance scale items that make up the SERVQUAL scale appears to be well supported both by the procedures used to develop the items and by their subsequent use as reported in the literature (cf. Carman 1990). We therefore conclude that these 22 performance items adequately define the domain of service quality and we use the same performance items to examine the proposed alternative to the SERVQUAL scale and in the analyses of the relationships between service quality, consumer satisfaction, and purchase intentions.

Research Models and Propositions

We investigate four specific questions that correspond to the three research steps identified in the Methods section. The first question is directed at the measurement of the service quality construct. Specifically, the ability of the more concise performance-only scale suggested by the literature review (SERVPERF, equation 6) is compared with that of three alternatives:

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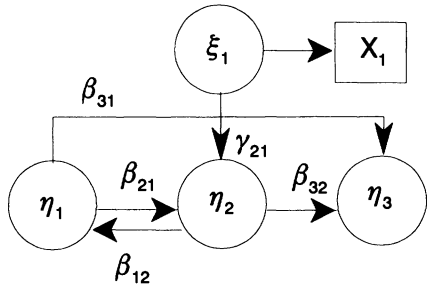
FIGURE 2
The Structural Models



MODEL 1

ξ_1 = SERVQUAL
η_1 = Consumer Satisfaction
η_2 = Overall Service Quality
η_3 = Purchase Intentions

MODEL 2

ξ_1 = SERVPERF
η_1 = Consumer Satisfaction
η_2 = Overall Service Quality
η_3 = Purchase Intentions

SERVQUAL (equation 4), weighted SERVQUAL (equation 5), and weighted SERVPERF (equation 7).

$$\text{Service Quality} = (\text{Performance} - \text{Expectations}) \quad (4)$$

$$\text{Service Quality} = \text{Importance}^* (\text{Performance} - \text{Expectations}) \quad (5)$$

$$\text{Service Quality} = (\text{Performance}) \quad (6)$$

$$\text{Service Quality} = \text{Importance}^* (\text{Performance}) \quad (7)$$

The first proposition provides the basis for our investigation:

P_1: An unweighted performance-based measure of service quality (unweighted SERVPERF) is a more appropriate basis for measuring service quality than SERVQUAL, weighted SERVQUAL, or weighted SERVPERF.

The evaluation P_1 calls for an assessment of whether the addition of the importance weights suggested by Zeithaml, Parasuraman, and Berry (1990) improves the ability of the SERVQUAL and SERVPERF scales to measure service quality and a direct comparison of the two measurement approaches. On the basis of the findings by Bolton and Drew (1991a), and the attitude and satisfaction literatures reviewed previously, the addition of importance weights is not expected to improve either scale and the SERVPERF alternative is expected to outperform the SERVQUAL scale.

The structural models identified in Figure 2 are used to further the consideration of the SERVQUAL and SERVPERF scales as well as to consider the three remaining research questions. As discussed in the literature review, the SERVPERF scale appears to conform more closely to the implications of the satisfaction and attitude literatures. Therefore, we propose that the model incorporating SERVPERF (model 2) will have a better fit (as measured by the chi square statistic and the measurement model's adjusted goodness of fit) because the performance-only form is more consistent with established theory (cf. Mazis, Ahtola, and Klippel 1975) and hence the SERVPERF measurement model should more closely approximate the theoretical model identified in Figure 2.

The second objective of our study is to examine the relationships between service quality, consumer satisfaction, and purchase intentions. The following three additional propositions identify the questions addressed in this part of the study.

P_2: Customer satisfaction is an antecedent of perceived service quality.

P_3: Consumer satisfaction has a significant impact on purchase intentions.

P_4: Perceived service quality has a significant impact on purchase intentions.

The first question considered is the causal order of the perceived service quality–satisfaction relationship (P_2). This analysis is also based on a consideration of the structural models identified in Figure 2. Specifically, P_2 proposes that the path (B_{21}) showing consumer satisfaction as an antecedent of service quality should have a statistically significant ($p \leq .05$) LISREL estimate whereas the estimate of the reverse path (satisfaction as an outcome of service quality, B_{12}) should not be significant (cf. Bitner 1990; Bolton and Drew 1991a,b).

The next question investigated is whether consumers' level of satisfaction with a service provider affects their purchase intentions toward that firm (P_3). Again, the structural models are used to investigate this proposition. Specifically, in models that confirm (i.e., the theoretical model is not rejected), the LISREL estimate for the path linking satisfaction and purchase intention (B_{31}) is examined to determine whether the effect is significant ($p \leq .05$).

The final question addressed is whether consumers' perceptions of service quality affect their purchase intentions (P_4). The investigation of this proposition is identical to that of P_3 but the path of interest is between service quality and purchase intentions (B_{32}).

Methods

Organization of the Research

Step 1: Examining the dimensionality of SERVQUAL. In this step, the confirmatory factor analysis capabilities of LISREL VII were used to determine whether the 22 items that define the SERVQUAL scale have the same factor loading pattern for the firms investi-

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gated as was found by Parasuraman, Zeithaml, and Berry (1988). To the extent that similar factor structures are identified (see Figure 1), evidence of the reliability of the SERVQUAL scale is produced. If the five-component structure is not confirmed, the OBLIMIN factor analysis procedure in SPSS-X and a reliability analysis can be used to assess the dimensionality and reliability of the items.

Step 2: Comparison of alternative measures of service quality. On the basis of the theoretical concerns discussed previously, we assessed three alternatives to the SERVQUAL scale. Specifically, in this step we examined the original SERVQUAL scale (equation 4), an importance-weighted SERVQUAL scale (equation 5), a performance-based approach to the measurement of service quality (SERVPERF, equation 6), and an importance-weighted version of the SERVPERF scale (equation 7). This examination proceeded in two stages. First, the ability of each of the four scales to explain variation in service quality was assessed by regressing the individual items comprising each of the alternative scales against a measure of the respondents' perceptions of the overall quality inherent in the services offered by the eight firms included in the sample (see Appendix, variable 85).

Second, each measure's theoretical support was examined in an analysis of the structural models identified in Figure 2. Specifically of interest were (1) the degree of fit of the respective models and (2) the significance of the effect on service quality attributed to each of the alternative measures (i.e., the significance of the path between the perceived service quality scale used and the overall measure of service quality, path γ_{21}).

Step 3: Analysis of relationship between service quality, consumer satisfaction, and purchase intention. The third step extended the research beyond the question of which approach to the measurement of service quality is the most appropriate. Here we considered (1) the causal order of the consumer satisfaction–service quality relationship, (2) the effect of consumer satisfaction on purchase intentions, and (3) the effect of service quality on purchase intentions. These relationships were also investigated through the analysis of the structural models identified in Figure 2. Specifically, we investigated each relationship by examining the significance of the LISREL VII estimated path coefficient that links the variables noted.

The Sample

Data were gathered from personal interviews conducted in a medium-sized city in the southeastern United States. A total of 660 usable questionnaires (all questions answered) were gathered randomly from consumers at their residences by trained interviewers during a two-week period in the summer of 1988. The sampling frame was the entire population of the city. Interviewers were instructed to solicit responses randomly and were assigned city areas to prevent overlap.

Responses were gathered on the service quality offered by two firms in each of four industries: banking, pest control, dry cleaning, and fast food. Because of the length of the questionnaire, respondents were asked to evaluate only one firm. The sample size for each industry was: banking 188 (firm 1, 92 and firm 2, 96); pest control 175 (firm 1, 91 and firm 2, 84); dry cleaning 178 (firm 1, 88 and firm 2, 90); fast food 189 (firm 1, 98 and firm 2, 91). The firms and industries were chosen on the basis of the results of a convenience survey suggesting that these were the four service industries most familiar to the area's consumers. The two firms chosen within each industry were those with the largest sales volume in the city where the sample was drawn (for the banks, the number of depositors was used to qualify the firms included). Respondents were screened to determine whether they had used one of the service providers included in the study within the last 30 days. This screening ensured that the respondents were familiar with the firm whose services they were asked to evaluate.

Measures

The measures needed for the study were expectations, perceptions of performance, and importance measures to construct the four alternative measures of service quality, a direct measure of service quality, a measure of consumer satisfaction, and a purchase intentions measure. The 22 expectation (see Appendix, variables E_1-E_{22}) and performance (see Appendix, variables P_1-P_{22}) items were taken directly from the SERVQUAL scale (Parasuraman, Zeithaml, and Berry 1988). The importance weights were adapted from the wording of the expectation and performance items included in the original SERVQUAL scale (see Appendix, variables I_1-I_{22}). The direct measure of service quality was based on responses to a 7-point semantic differential question (see Appendix, variable 85). In addition, self-report measures of consumer satisfaction and purchase intentions were constructed similarly (see Appendix, variables 87 and 84, respectively).

Results

Dimensionality, Reliability, and Validity of Service Quality Measures (Step 1)

Dimensionality and reliability. The first step was to examine the dimensionality of the current service quality

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scale (SERVQUAL) by means of a confirmatory factor analysis. Table 1 gives the results of the LISREL VII–based analysis for each of the four types of service firms (banks, pest control, dry cleaning, and fast food). These results suggest that the 5-component structure proposed by Parasuraman, Zeithaml, and Berry (1988) for their SERVQUAL scale (see Figure 1) is not confirmed in any of the research samples. Specifically, the chi square statistic universally indicates a poor fit between the theoretical and measurement models for the 5-component structure. The adjusted goodness-of-fit indices (AGFI) are also not indicative of a good fit as they range from .740 to .831.

Because the 5-factor structure was not confirmed, we decided to assess the unidimensionality of the 22 items. We performed a factor analysis of the SERVQUAL and SERVPERF scales using the OBLIMIN oblique factor rotation procedure in SPSS-X. All of the items loaded predictably on a single factor with the exception of item 19 (see Table 2), which loaded very weakly in the analysis of the SERVQUAL scale and had a negative loading for both scales. It was therefore dropped and coefficient alpha for both scales and all subsamples (each industry) was recalculated. As is indicated in Table 2, the reliability in every case (coefficient alpha in excess of .800) suggests that both scales can be treated as unidimensional. Thus, in the analysis that follows, the 21 retained items are either summed or averaged (to develop the four service quality scales in the LISREL VII analysis of the structural models) or they are considered as one composite set of individual measures (in the stepwise regression analysis).

Validity. The primary threat to the validity of the measures used in this study is construct validity. Carmines and Zeller (1979, p. 23) state, "[f]undamentally, construct validity is concerned with the extent to which a particular measure relates to other measures consistent with theoretically derived hypotheses concerning the concepts (or constructs) that are being measured." They further suggest that the process of construct validation is by definition theory-laden. Churchill (1979) suggests that convergent and discriminant validity should be assessed in investigations of construct validity. Convergent validity involves the extent to which a measure correlates highly with other measures designed to measure the same construct. Therefore, we examined a correlation matrix of all the items tested in models 1 and 2 (see Table 3). A high correlation between the items SERVPERF, importance-weighted SERVPERF, and service quality indicates some degree of convergent validity. Discriminant validity involves the extent to which a measure is novel and does not simply reflect some other variable. Churchill (1979) suggests assessing discriminant validity by determining whether the correlation between two different measures of the same variable is higher than the correlation between the measure of that variable and those of any other variable. Again, an examination of the correlation matrix in Table 3 indicates discriminant validity of the research variables as the three service quality scales all correlate more highly with each other than they do with other research variables (i.e., satisfaction and purchase intentions). Hence, we suggest that the proposed performance-based measures provide a more construct-valid explication of service quality because of their content validity (i.e., use of importance weights and use of performance-based measures are arguably more theoretically sound approaches) and the evidence of their discriminant validity.

TABLE 1
Confirmatory Factor Analysis Parameter Estimates for 5-Factor Conceptualization of Service Quality

Parameter	Banks	Pest Control	Dry Cleaning	Fast Food
Chi square	308.60	486.16	402.60	364.16
d.f.	204	204	204	204
p	.000	.000	.000	.000
GFI[a]	.863	.790	.819	.849
AGFI[b]	.831	.740	.776	.813
RMSR[c]	.309	.466	.381	.515

[a]Goodness of fit.
[b]Adjusted goodness of fit.
[c]Root mean square residual.

Comparison of Alternative Measures of Service Quality (Step 2)

P_1 suggests that the unweighted SERVPERF scale should capture more of the variation in service quality than any of the other identified alternatives (SERVQUAL, weighted SERVQUAL, and weighted SERVPERF). The stepwise regression analysis summarized in Table 4 affirms P_1. In all of the four service industries examined, unweighted SERVPERF explains more of the variation in the global measure of service quality (see Table 4).

In addition, a comparison of the SERVQUAL and weighted SERVQUAL scales (columns 1–4 and 5–8 of Table 4) indicates that the unweighted SERVQUAL scale explains more of the variation in service quality in three of the four industries (the exception being dry cleaning). We therefore decided to use only the unweighted SERVPERF and SERVQUAL scales in the structural analysis of the relationships between these scales, service quality, consumer satisfaction, and purchase intentions because they arguably represent

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TABLE 2
Factor Analysis of 22 Individual Dimensions of Service Quality

	SERVQUAL				SERVPERF			
Variable	Banks	Pest Control	Dry Cleaning	Fast Food	Banks	Pest Control	Dry Cleaning	Fast Food
V1	.396	.697	.577	.181	.480	.820	.692	.408
V2	.397	.368	.492	.249	.463	.652	.614	.458
V3	.477	.523	.536	.339	.557	.842	.642	.499
V4	.381	.319	.398	.055	.485	.703	.640	.384
V5	.781	.741	.736	.543	.804	.831	.774	.572
V6	.728	.753	.798	.543	.726	.828	.760	.683
V7	.826	.837	.805	.748	.822	.891	.856	.669
V8	.791	.832	.789	.679	.799	.873	.785	.679
V9	.833	.694	.654	.380	.788	.835	.626	.349
V10	.346	.467	.209	.325	.355	.532	.281	.136
V11	.568	.611	.358	.657	.640	.712	.483	.607
V12	.522	.622	.499	.706	.631	.706	.539	.672
V13	.500	.556	.392	.706	.623	.789	.538	.660
V14	.572	.622	.730	.409	.685	.785	.771	.550
V15	.817	.676	.762	.595	.815	.788	.836	.665
V16	.573	.764	.740	.641	.638	.793	.803	.689
V17	.647	.608	.673	.544	.688	.702	.666	.518
V18	.535	.563	.472	.412	.620	.762	.483	.429
V19	−.337	−.298	−.165	.027	−.677	−.769	−.615	−.474
V20	.459	.502	.399	.422	.580	.685	.490	.485
V21	.502	.571	.522	.464	.552	.670	.703	.573
V22	.272	.420	.399	.156	.345	.598	.403	.280
Eigenvalue	7.472	8.229	7.437	5.194	9.037	12.651	9.378	6.408
% of variation	34.8%	37.4%	33.8%	23.6%	41.1%	57.5%	42.6%	29.1%
Coefficient alpha[a]	.890	.901	.900	.849	.925	.964	.932	.884

[a]Item V19 excluded.

TABLE 3
Correlation Coefficients for Structural Models in Figure 2

	SERVQUAL	Weighted SERVQ-UAL	SERVPERF	Weighted SERVP-ERF	Overall service quality	Satisfaction	Purchase intention
SERVQUAL	1.0000						
Weighted SERVQUAL	.9787	1.0000					
SERVPERF	.8100	.7968	1.0000				
Weighted SERVPERF	.6589	.6307	.9093	1.0000			
Overall service quality	.5430	.5394	.6012	.5572	1.0000		
Satisfaction	.5605	.5559	.5978	.5513	.8175	1.0000	
Purchase intention	.3534	.3613	.3647	.3486	.5272	.5334	1.0000

the best of each of the two alternative conceptualizations of service quality.

Relationships Between Service Quality, Customer Satisfaction, and Purchase Intentions (Step 3)

Figure 2 identifies the two models used to investigate P_2, P_3, and P_4 and to further the comparison of the performance- and disconfirmation-based measures of service quality (SERVPERF and SERVQUAL, respectively). Models 1 and 2 are identical with the exception that SERVQUAL is used to measure service quality in model 1 whereas SERVPERF is used in model 2. The models conceptualize a nonrecursive ("two-way") relationship between service quality and satisfaction in order to test simultaneously the effects hypothesized by Parasuraman, Zeithaml, and Berry (1985, 1988) (service quality is an antecedent of customer satisfaction) and by Bitner (1990) and Bolton

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TABLE 4
The Variation Explained by the Alternative Measures of Service Quality

	SERVQUAL				Weighted SERVQUAL				SERVPERF				Weighted SERVPERF			
	Banking	Pest Control	Dry Cleaning	Fast Food	Banking	Pest Control	Dry Cleaning	Fast Food	Banking	Pest Control	Dry Cleaning	Fast Food	Banking	Pest Control	Dry Cleaning	Fast Food
V1[1]																
V2								.164[b]				.157[b]				.256[c]
V3				.143[a]												
V4	.147[a]								.248[c]							
V5					.194[a]	−.227[a]										
V6	.284[c]	−.200[a]	.307[c]	.255[c]	.222[b]	−.186[a]	.277[b]	.282[c]	.350[c]		.380[c]		.267[b]		.240[b]	
V7	.478[c]				.452[c]		.268[b]	.234[b]	.351[b]			.407[c]	.437[c]		.242[b]	.323[c]
V8		.594[c]				.614[c]				.304[b]				.346[c]		
V9	−.216[a]				−.248[b]			−.131[a]	−.231[b]				−.195[a]			
V10														.193[b]		
V11			.158[a]				.268[c]			.256[c]						
V12													.253[c]			.242[c]
V13				.141[a]								.152[a]				.194[b]
V14		.352[c]				.329[c]		−.153[a]								.130[a]
V15											.191[a]					
V16			.235[b]	.187[b]				.318[c]		.175[a]	.349[c]	.165[a]			.189[a]	
V17											−.219[b]			.161[a]		
V18																
V19			−.127[a]	−.163[a]			−.135[a]								−.164[b]	
V20									.158[b]			.141[a]	.181[b]			
V21	.189[b]				.167[b]											
V22		1.42				.157[a]										
R^2	.46511[2]	.36515	.30747	.41534	.44813	.36316	.36958	.38332	.47895	.38760	.44675	.47585	.40333	.33726	.43166	.46718

Where: a = $p < .05$.
b = $p < .01$.
c = $p < .001$.
[1]VI to V22 are the alternative scale items of service quality (see Appendix A, variables P1 to P22). Entries in the cells represent correlation coefficients. All nonsignificant coefficients are ommitted.
[2]Numbers in each cell are adjusted R^2s.

and Drew (1991a,b) (service quality is an outcome of customer satisfaction). In addition, the model suggests that both service quality and satisfaction affect purchase intentions.

However, before considering P_2, P_3, and P_4, we assessed the fit of the two respective models to the data (see Table 5). Model 1 (SERVQUAL) had a good fit in two of the four industries (banking and fast food) whereas model 2 (SERVPERF) had an excellent fit in all four industries. Because the only difference in the two models is the measure of service quality used (either SERVQUAL or SERVPERF), these results were interpreted as additional support for the superiority of the SERVPERF approach to the measurement of service quality.

Because of this superiority, we used model 2 to assess the strength of the relationships between service quality, consumer satisfaction, and purchase intention. This analysis suggests that (1) service quality has a significant ($p \leq .05$) effect on consumer satisfaction in all four samples (see Table 5, model 2, path β_{12}), (2) consumer satisfaction has a significant ($p \leq .05$) effect on purchase intentions in all four samples (see Table 5, model 2, path β_{31}), and (3) service quality does not have a significant ($p \leq .05$) impact on purchase intentions in any of the samples (see Table 5, model 2, path β_{32}). Thus, P_2 and P_3 both receive strong support from the results, though the direction of the effect observed in the consideration of P_2 is the opposite of that proposed. The analysis of P_4 afforded no support for the proposed effect.

Discussion

We investigated three main questions:

- How should service quality be conceptualized and measured?
- What is the causal order of the relationship between service quality and consumer satisfaction?
- What impacts do service quality and consumer satisfaction have on purchase intentions?

In answer to the first question, the literature review and empirical results both suggest that service quality should be conceptualized and measured as an attitude. The literature clearly supports the performance-only (SERVPERF) approach. In the empirical analysis, the first step calls into question the efficacy of the 5-component conceptualization of service quality offered by Parasuraman, Zeithaml, and Berry

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TABLE 5
Standardized Parameter Estimates for Causal Models

		Banking		Pest Control		Dry Cleaning		Fast Food	
	Parameter	LISREL Estimate	T-Value	LISREL Estimate	T-Value	LISREL Estimate	T-Value	LISREL Estimate	T-Value
SERVQUAL Model (1)									
	B_{21}	−1.796	−1.512	−2.810	−1.341	−8.398	−.408	−.055	−.300
	B_{12}	1.113	14.794	1.099	10.620	1.103	15.256	.904	10.566
	B_{31}	.668	3.712	.646	4.247	.836	4.598	.343	2.774
	B_{32}	.280	1.475	.301	2.033	.099	.542	.296	2.187
	γ_{21}	2.417	2.226	2.289	1.746	7.157	.452	.812	5.235
	Chi square	.000	(p = .972)	5.090	(p = .024)	4.060	(p = .044)	6.020	(p = .140)
	AGFI	1.000		.863		.890		.838	
	RMS	.001		.068		.061		.063	
	SMC-Y_1	.664		.465		.750		.647	
	-Y_2	−3.482		−7.309		−58.200		.254	
	-Y_3	.325		.326		.409		.260	
SERVPERF Model (2)									
	B_{21}	−1.353	−1.595	−1.377	−1.944	−2.904	−.989	−.141	−.845
	B_{12}	1.109	14.156	1.006	11.793	1.065	17.584	.944	12.787
	B_{31}	.550	3.124	.659	4.323	.837	4.598	.362	2.924
	B_{32}	.374	1.979	.285	1.926	.098	.535	.282	2.069
	γ_{21}	2.154	2.585	1.683	3.202	3.644	1.300	1.179	6.122
	Chi square	.080	(p = .781)	.220	(p = .639)	3.290	(p = .070)	.230	(p = .629)
	AGFI	.998		.994		.910		.994	
	RMS	.009		.012		.044		.010	
	SMC-Y_1	.657		.521		.768		.652	
	-Y_2	−2.298		−1.886		−7.799		.278	
	-Y_3	.305		.325		.412		.266	

(1988). The second step indicates that the SERVPERF scale explains more of the variation in service quality than does SERVQUAL. Both the literature review and the analysis of the structural models (see Figure 2 and Table 5, models 1 and 2) suggest that the SERVQUAL conceptualization is in fact flawed: (1) it is based on a satisfaction paradigm rather that an attitude model and (2) the empirical analysis of the structural model suggests that the SERVQUAL model (see Figure 2 and Table 5, model 1) confirms in only two of the four industries. Thus, the weight of the evidence clearly supports the use of performance-based measures of service quality.

The second question investigated is the causal order of the satisfaction–service quality relationship. Much of the recent literature has suggested that satisfaction is an antecedent of service quality (cf. Bitner 1990; Bolton and Drew 1991a,b). However, the analysis of the research model indicates that this may not be the case and provides empirical support for the notion that perceived service quality in fact leads to satisfaction as proposed by Parasuraman, Zeithaml, and Berry (1985, 1988).

The third question pertains to the effects of service quality and satisfaction on purchase intentions (see Figure 2 and Table 5). The analysis of the LISREL estimates (model 2; see Table 5) suggests that satisfaction has a significant ($p \leq .05$) effect on purchase intentions in all four samples whereas service quality does not have such an effect in any of the four industries. From the significance tests summarized in Table 5, satisfaction appears to have a stronger and more consistent effect on purchase intentions than does service quality.

Conclusions and Managerial Implications

The major conclusion from our study is that marketing's current conceptualization and measurement of service quality are based on a flawed paradigm. We present empirical and literature support suggesting that service quality should be measured as an attitude. The performance-based scale developed (SERVPERF) is efficient in comparison with the SERVQUAL scale; it reduces by 50% the number of items that must be measured (44 items to 22 items). The analysis of the structural models also supports the theoretical superiority of the SERVPERF scale (see Table 5); only the model that uses the SERVPERF scale consistently confirmed (model 2). These factors, along with the failure of the 5-component SERVQUAL model to confirm (see Table 1), support the use of a performance-based measure of service quality.

The remaining questions addressed in our study

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are essential managerial issues. The results suggest that service quality is an antecedent of consumer satisfaction and that consumer satisfaction exerts a stronger influence on purchase intentions than does service quality. Thus, managers may need to emphasize total customer satisfaction programs over strategies centering solely on service quality. Perhaps consumers do not necessarily buy the highest quality service; convenience, price, or availability may enhance satisfaction while not actually affecting consumers' perceptions of service quality.

Finally (see Table 4), the results from step 1 also suggest that the scale items that define service quality in one industry may be different in another. Perhaps high involvement services such as health care or financial services have different service quality definitions than low involvement services such as fast food or dry cleaning. Managers and researchers therefore must consider the individual dimensions of service quality when making cross-sectional comparisons. Managers should also be able to adjust their marketing strategies more effectively when the full set of individual scale items are considered.

Implications for Future Research

Our research has only begun to address the many issues that are important in the management of services. The findings undoubtedly raise more questions than they answer, but the questions we address—how the service quality construct should be measured and how service quality is related to consumer satisfaction and purchase intentions—are arguably among the most important concerns in services marketing.

Future studies should consider other attitude-based conceptualizations and extend beyond the four service industries sampled in our study. The nature of the relationship between consumer satisfaction and service quality appears to be an area in great need of additional exploration. Investigations of the roles of satisfaction and service quality in predicting market share also appear well directed and may enhance our understanding of the role of these constructs in the formation of purchase intentions. The failure of service quality to affect purchase intentions consistently should be a concern for both managers and researchers. Perhaps consumers do not always buy the best quality service. Might they instead purchase on the basis of their assessment of the value of a service? Perhaps future research should develop measures of service performance that utilize other criteria, such as value, for determining whether a service is purchased. Finally, our study was specific to a service context. Generalizing the results to goods industries may not be possible. The ever-increasing magnitude of the service sector simply underscores the need for managers and researchers alike to increase the attention directed at the important issues in the marketing of services.

Limitations

In designing our study, we attempted to minimize its limitations. However, generalizations beyond the four specific service industries investigated are tenuous. Future studies should incorporate multiple measures of all of the constructs examined. Limiting the study to the two highest market share firms in each category may also have affected the variable distributions and, hence, the importance of the predictors. In addition, with the possible exception of banking, the services investigated are all low involvement service categories. Perceived quality may play a bigger role (in comparison with satisfaction) in high involvement situations, where a firm may need to do more than simply meet customers' "minimum requirements." Finally, the number of constructs other researchers could add to the models examined is probably unlimited.

Appendix

Expectations

This survey deals with **your opinions of ____ services**. Please show the extent to which you think institutions offering telephone services should possess the features described in each statement. Do this by using the scale presented below. If you strongly agree that these institutions should possess a feature, place a seven on the line preceding the statement. If you strongly disagree that these institutions should possess a feature, place a one on the line. If your feelings are not strong, place one of the numbers between one and seven on the line to properly reflect the actual strength of your feelings. There are no right or wrong answers—all we are interested in is a number that best shows your Expectations about institutions offering banking services.

1 - - - 2 - - - 3 - - - 4 - - - 5 - - - 6 - - - 7

STRONGLY DISAGREE — STRONGLY AGREE

____ E1. They should have up-to-date equipment & technology.

____ E2. Their physical facilities should be visually appealing.

____ E3. Their employees should be well dressed and appear neat.

____ E4. The appearance of the physical facilities of these institutions should be in keeping with the type of services provided.

____ E5. When these institutions promise to do something by a certain time, they should do so.

____ E6. When customers have problems, these institutions should be sympathetic and reassuring.

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____ E7. These institutions should be dependable.

____ E8. They should provide their services at the time they promise to do so.

____ E9. They should keep their records accurately.

____E10. They shouldn't be expected to tell their customers exactly when services will be performed.

____E11. It is not realistic for customers to expect prompt service from employees of these institutions.

____E12. Their employees don't always have to be willing to help customers.

____E13. It is okay if they are too busy to respond to customer requests promptly.

____E14. Customer should be able to trust employees of these institutions.

____E15. Customers should be able to feel safe in their transactions with these institutions' employees.

____E16. Their employees should be polite.

____E17. Their employees should get adequate support from these institutions to do their jobs well.

____E18. These institutions should not be expected to give customers individual attention.

____E19. Employees of these institutions cannot be expected to give customers personal attention.

____E20. It is unrealistic to expect employess to know what the needs of their customers are.

____E21. It is unrealistic to expect these institutions to have their customers' best interests at heart.

____E22. They shouldn't be expected to have operating hours convenient to all their customers.

Performance

The following set of statements relate to **your feelings about XYZ ____**. For each statement, please show the extent to which you believe XYZ ____ has the feature described by the statement. Once again, placing a seven on the line means you strongly agree that XYZ has that feature, and a one means you strongly disagree. You may use any of the numbers in the middle as well to show how strong your feelings are. There are no right or wrong answers—all we are interested in is a number that best shows your perceptions about XYZ whether you use their service or not.

1 - - - 2 - - - 3 - - - 4 - - - 5 - - - 6 - - - 7

STRONGLY DISAGREE	STRONGLY AGREE

____ P1. XYZ____has up-to-date equipment.

____ P2. XYZ____'s physical facilities are visually appealing.

____ P3. XYZ____'s employees are well dressed and appear neat.

____ P4. The appearance of the physical facilities of XYZ____is in keeping with the type of service provided.

____ P5. When XYZ____promises to do something by a certain time, it does so.

____ P6. When you have problems, XYZ____is sympathetic and reassuring.

____ P7. XYZ____is dependable.

____ P8. XYZ____provides its services at the time it promises to do so.

____ P9. XYZ____keeps its records accurately.

____P10. XYZ____does not tell its customers exactly when services will be performed.

____P11. You do not receive prompt service from XYZ ____employees.

____P12. Employees of XYZ____are not always willing to help customers.

____P13. Employees of XYZ____are too busy to respond to customer requests promptly.

____P14. You can trust employees of XYZ____.

____P15. You can feel safe in your transactions with XYZ ____'s employees.

____P16. Employees of XYZ____are polite.

____P17. Employees get adequate support from XYZ ____to do their jobs well.

____P18. XYZ____does not give you individual attention.

____P19. Employees of XYZ____do not give you personal attention.

____P20. Employees of XYZ____do not know what your needs are.

____P21. XYZ____does not have your best interests at heart.

____P22. XYZ____does not have operating hours convenient to all their customers.

Importance

The following set of statements relate to **your feelings about the importance of each feature described in your decision to purchase ____ services.** A seven means you consider the feature very important in deciding where to purchase banking services, a one means it is very unimportant. You may place any of the numbers shown on the scale below beside each feature to indicate its importance to you. There are no right or wrong answers—all we are interested in is your perception of how important each feature is to you in your decision where to purchase banking services.

1 - - - 2 - - - 3 - - - 4 - - - 5 - - - 6 - - - 7

VERY UNIMPORTANT	VERY IMPORTANT

____ I1. Up-to-date equipment.

____ I2. Physical facilities that are visually appealing.

____ I3. Employees that are well dressed and appear neat.

____ I4. Physical facilities that appear to be in keeping with the type of service provided.

____ I5. When something is promised by a certain time, doing it.

____ I6. When there is a problem, being sympathetic and reassuring.

____ I7. Dependability.

____ I8. Providing service by the time promised.

____ I9. Accurate record keeping.

____I10. Telling the customer exactly when the service will be performed.

____I11. Receiving prompt service.

____I12. Employees who are always willing to help customers.

____I13. Employees who are not too busy to respond to customer request promptly.

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____I14. Employees who are trustworthy.
____I15. The feeling that you are safe when conducting transactions with the firm's employees.
____I16. Employees who are polite.
____I17. Adequate support from the firm so employees can do their job well.
____I18. Individual attention.
____I19. Employees who give you personal attention.
____I20. Employees who know what your needs are.
____I21. A firm which has your best interests at heart.
____I22. Convenient operating hours.

Other Measures

The following set of statements relate to **your feelings about XYZ** ____. Please respond by circling the number which best reflects your own perceptions.

(Future Purchase Behavior)

(84) In the next year, my use of XYZ _____ will be

1 - - - 2 - - - 3 - - - 4 - - - 5 - - - 6 - - - 7
NOT AT ALL VERY FREQUENT

(Overall Quality)

(85) The quality of XYZ ____'s services is

1 - - - 2 - - - 3 - - - 4 - - - 5 - - - 6 - - - 7
VERY POOR EXCELLENT

(Satisfaction)

(87) My feelings towards XYZ ____'s services can best be described as

1 - - - 2 - - - 3 - - - 4 - - - 5 - - - 6 - - - 7
VERY UNSATISFIED VERY SATISFIED

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